# Master of Data Analytics

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## Multiclass Classification on the UCI Seeds Dataset Using a Feed-Forward Neural Network

## Introduction

This report summarizes my work on building and evaluating a feed-forward Artificial Neural Network (ANN) for classifying wheat seed varieties using the UCI Seeds dataset. The dataset consists of 210 samples, each described by seven numeric features, and the goal is to predict one of three wheat classes. I followed the full machine learning workflow, including preprocessing, model design, hyperparameter comparison, and evaluation, keeping the writing and explanations in a simple, student-friendly tone while following APA-style structure.

# Methods

I loaded the dataset, assigned the required column names, and performed exploratory checks such as class distribution, dataset shape, and feature statistics. No missing values were present. The data was split into 80% training and 20% testing using stratified sampling with a fixed random seed for reproducibility. Features were standardized using StandardScaler, and labels were converted to one-hot vectors.

The ANN architecture consisted of two hidden layers with ReLU activation and L2 regularization. Two model variants were created: a smaller network (32 and 16 units) and a larger network (64 and 32 units). Both were trained for 100 epochs using the Adam optimizer. The purpose of testing two variants was to understand how hidden layer size affects performance.

## Results

The table below shows the performance of both ANN variants on the test set, using accuracy, precision, recall, and F1-score. The larger network performed slightly better overall.

| Model | Accuracy | Precision | Recall | F1-score |
|---|---|---|---|---|
| small_hidden | 0.8333 | 0.8416 | 0.8333 | 0.8218 |
| large_hidden | 0.8571 | 0.8634 | 0.8571 | 0.8499 |

Figure 1. Training and validation accuracy curves:

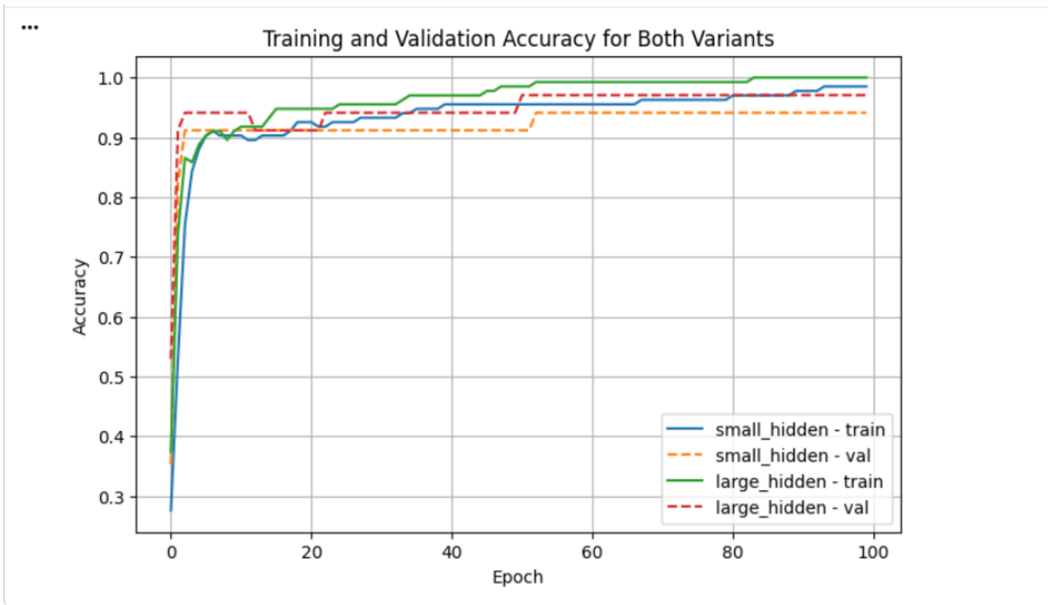
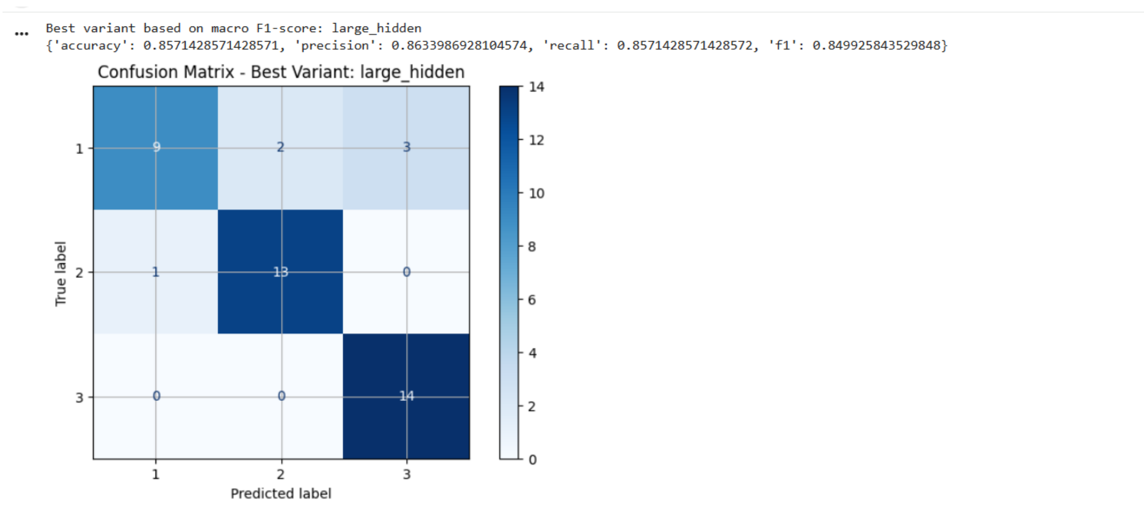



Figure 2. Confusion matrix of the best-performing model (large_hidden):



## Discussion

The larger neural network consistently performed better than the smaller version. Its higher accuracy and macro-level F1-score suggest that additional hidden units helped the model capture subtle differences among the three wheat classes. The confusion matrix shows that most predictions were correct, with a few mistakes mainly in class boundaries that are naturally close in feature space. A possible future improvement would be to test dropout, tune learning rates, or apply k-fold cross-validation for more stable estimates.

## AI Tool Use Disclosure

I learned and implemented most parts of this project independently by reviewing documentation, experimenting with model architectures, and testing different preprocessing steps. After completing the core implementation, I used AI tools such as ChatGPT only to clarify certain concepts, refine my report wording, and help structure the final document. All coding, model decisions, executions, and result interpretations were performed by me.